

## Pre-work · Axial Skeleton

Axial Skeleton, the skull and the spine. Read both Part A chapters first. There are more names to learn here than anywhere else in the course. Organizing them is what makes it manageable.

Module 2 · 2 of 4

Bring this to class

### How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

**1 Start with the reading.**

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

**2 Organize what you read.**

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

**3 Draw it.**

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

**4 Apply it.**

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

**5 Come back to it later in the week**

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

**Here is why the order works.** Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

### Organize it

2 min

PANEL 1 OF 3

Turn the skull and spine chapters into mini visuals. Eight chunks. The grids do the heavy lifting here.

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#### **YOU ONLY NEED FIVE SHAPES**

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

#### **Your chunks for this packet**

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

## 1 SPLIT

### Cranial against facial bones

One split, eight on one branch and fourteen on the other. Mark which are single and which are paired. This is the frame for the whole skull.

## 2 GRID

### The four sutures

Suture down the side, bones joined and location across the top. Add the pterion as a fifth row and mark why it matters.

## 3 GRID

### The major foramina

Foramen down the side, bone and what passes through across the top. Nine rows. Star the four you will confuse.

## 4 HUB

### The sphenoid

Sphenoid at the hub, because it touches every other cranial bone. Spoke to body, sella turcica, greater and lesser wings, pterygoid processes, optic canal, superior orbital fissure.

## 5 GRID

### The four fontanelles

Fontanelle down the side, location and closing age across the top. Four rows, and the ages are the part people lose.

## 6 LADDER

### The five regions of the spine

Five rungs, cervical at the top. Write the count beside each rung and the two fused regions at the bottom.

## 7 GRID

### Cervical, thoracic, lumbar vertebrae

Region down the side. Body shape, spinous process, and the one giveaway feature across the top. Three rows. The giveaway column is what lets you place a loose vertebra.

## 8 CHAIN

### The four curvatures, in developmental order

Thoracic and sacral present at birth, then cervical when the head lifts, then lumbar when the child walks. Arrows carry the milestones.

#### YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

## Draw it

2 min

### PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

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#### HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

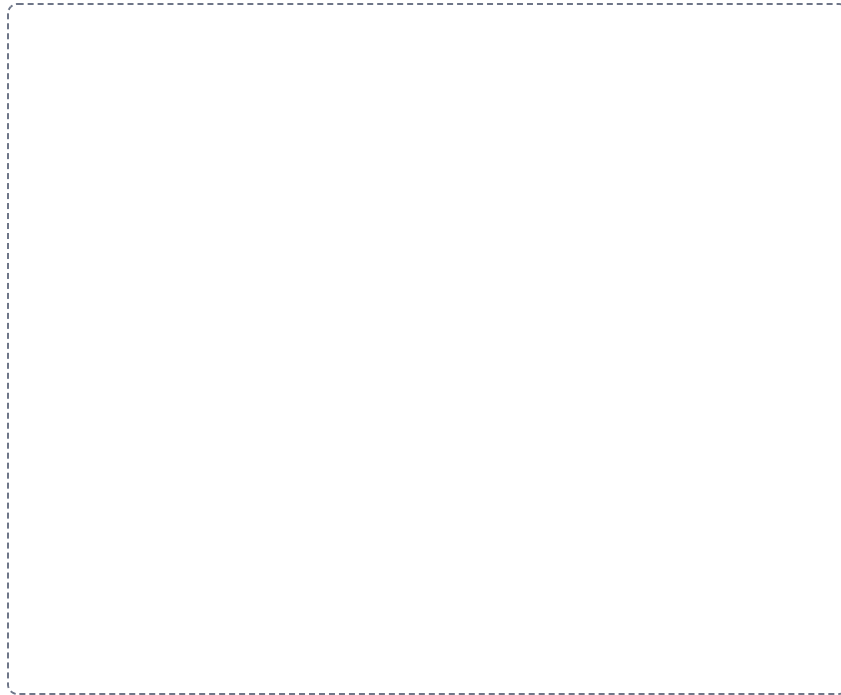
## DRAWING 1

### The skull, lateral view

#### BRAIN DUMP FIRST

Two minutes. Every bone you can name on the side of the skull, and the sutures between them.

Draw the skull in profile large enough to label. Outline each bone visible from the side and shade them differently: frontal, parietal, temporal, occipital, sphenoid, zygomatic, maxilla, nasal, lacrimal, mandible. Draw the four major sutures. Mark the pterion where four bones meet, the mastoid and styloid processes, the external acoustic meatus, and the zygomatic arch.



In your notebook, head the page **M2-4 The skull, lateral view** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can put your finger on the pterion and say which four bones meet there and what runs beneath it.

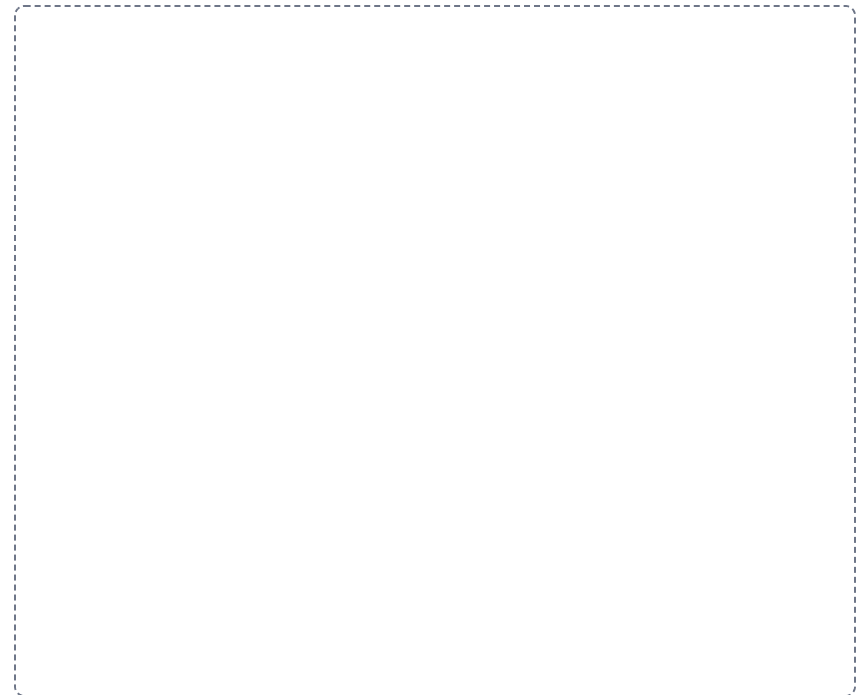
## DRAWING 2

### A typical vertebra, from above

#### BRAIN DUMP FIRST

One minute. Every part of a vertebra, and which region it belongs to.

Draw one vertebra looking down on it. Body at the front, then the pedicles, laminae, and the arch closing behind, with the vertebral foramen in the middle. Add the spinous process pointing back and the transverse processes to the sides, then the superior articular processes. Now redraw it twice more in a second color, once as cervical with transverse foramina and once as thoracic with costal facets.



In your notebook, head the page **M2-5 A typical vertebra, from above** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

Someone hands you a loose vertebra and you can name its region from one feature without hesitating.

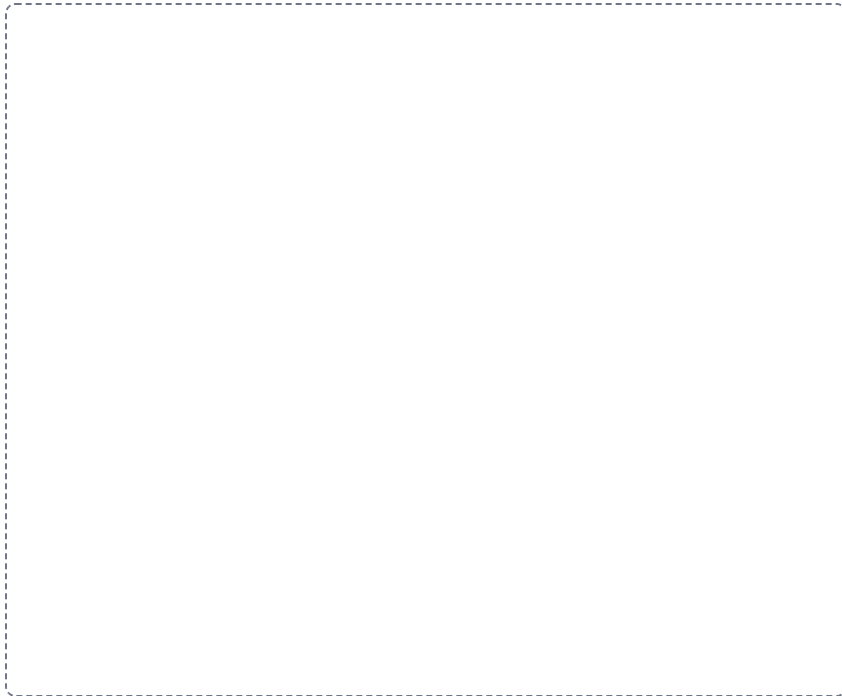
### DRAWING 3

#### The thoracic cage

##### BRAIN DUMP FIRST

One minute. The ribs, how they attach, and the parts of the sternum.

Draw the cage from the front. Sternum in three parts with the jugular notch, sternal angle, and xiphoid process labelled. Draw the twelve rib pairs and their costal cartilages, then bracket them: true, false, and floating. Mark the sternal angle at the level of the second rib.



In your notebook, head the page **M2-6 The thoracic cage** so you can find it later. Correct it in a second color.

##### YOU KNOW IT WHEN

You can count ribs from the sternal angle the way a clinician does, and say where you would stop for each group.

## Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

#### WORKED EXAMPLE

**A loose vertebra has a transverse foramen on each transverse process. Name the region and say how you know.**

**Answer.** Cervical.

**Reasoning.** Go to the giveaway feature first, not the size or the shape. Only cervical vertebrae have transverse foramina, because only they carry the vertebral arteries up to the brain. Thoracic vertebrae are placed by costal facets, and lumbar by their heavy body and blunt spinous process. So one feature settles it, and the other features are only confirmation.

#### Set A - Name the bone and the opening

1. The cribriform plate belongs to the \_\_\_\_\_ and passes \_\_\_\_\_.
2. The pituitary gland sits in the \_\_\_\_\_ of the \_\_\_\_\_.
3. The foramen magnum passes the \_\_\_\_\_ and the \_\_\_\_\_.
4. The only freely movable bone of the skull is the \_\_\_\_\_.
5. Name the seven bones that build the orbit.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

#### Set B - Predict it

1. A blow lands on the pterion. Predict the injury and name the vessel involved.  
\_\_\_\_\_  
\_\_\_\_\_
2. An infant has a bulging anterior fontanelle. Say what it may signal, and what a sunken one may signal instead.  
\_\_\_\_\_  
\_\_\_\_\_
3. A tooth infection in the upper jaw spreads. Name the cavity most at risk and say why.  
\_\_\_\_\_  
\_\_\_\_\_
4. A patient has a herniated disc at L4 to L5 with pain travelling down the leg. Name the two parts of the disc and explain the path of the pain.  
\_\_\_\_\_  
\_\_\_\_\_

#### Set C - Sort and classify

1. Sort as true, false, or floating: ribs 3, 9, 11.  
\_\_\_\_\_  
\_\_\_\_\_
2. Name the four curvatures and mark each primary or secondary, with the milestone that produces the secondary ones.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
3. Compare the atlanto-occipital and atlanto-axial joints by the movement each produces.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

#### YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.



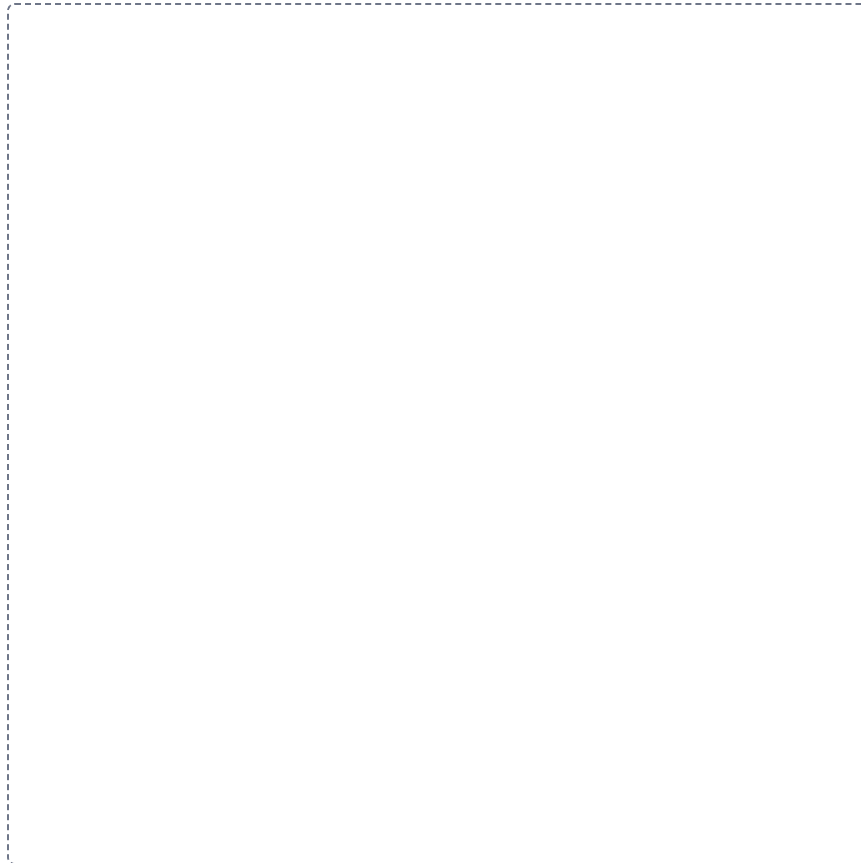
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M2-

### Cranial against facial bones

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

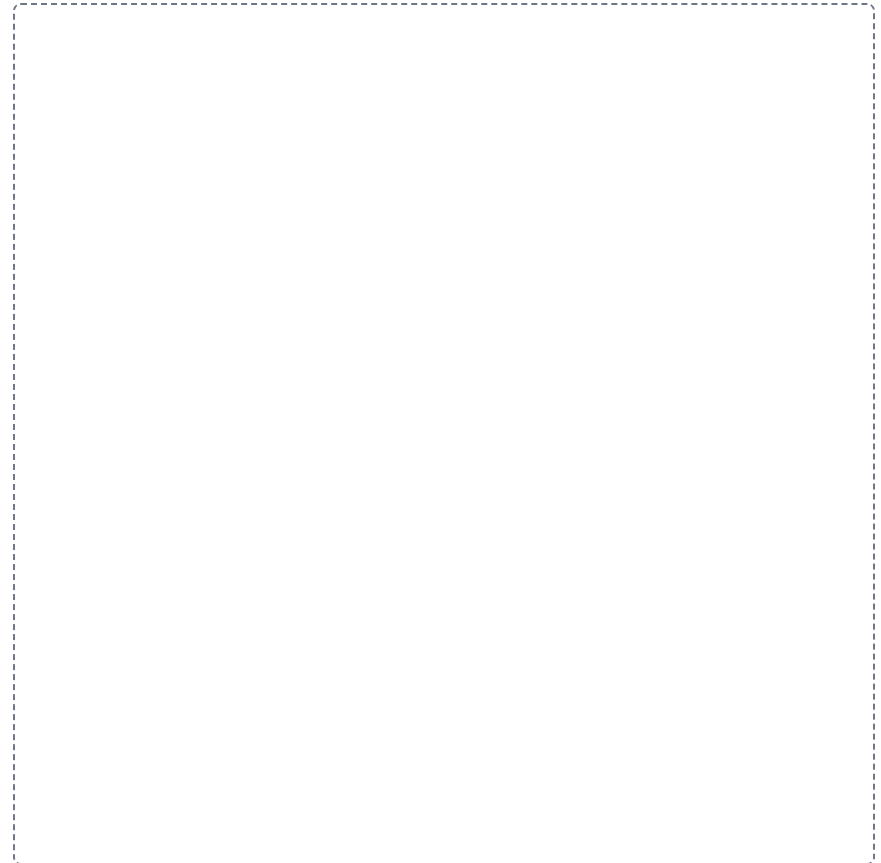


M2-

### The four sutures

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

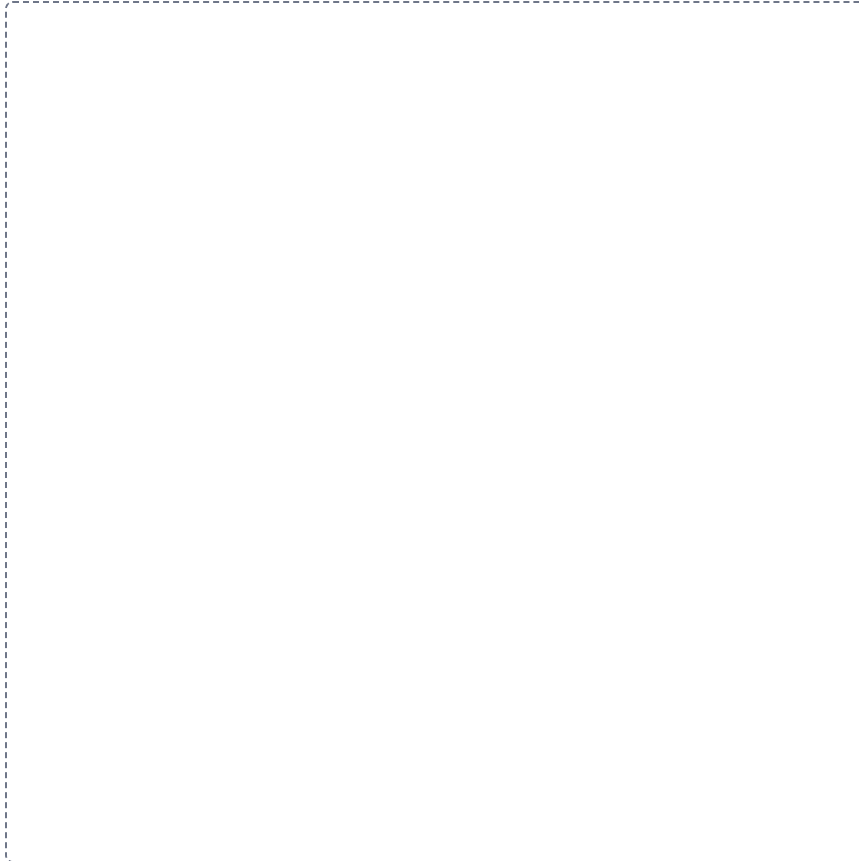


12-3

## The major foramina

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

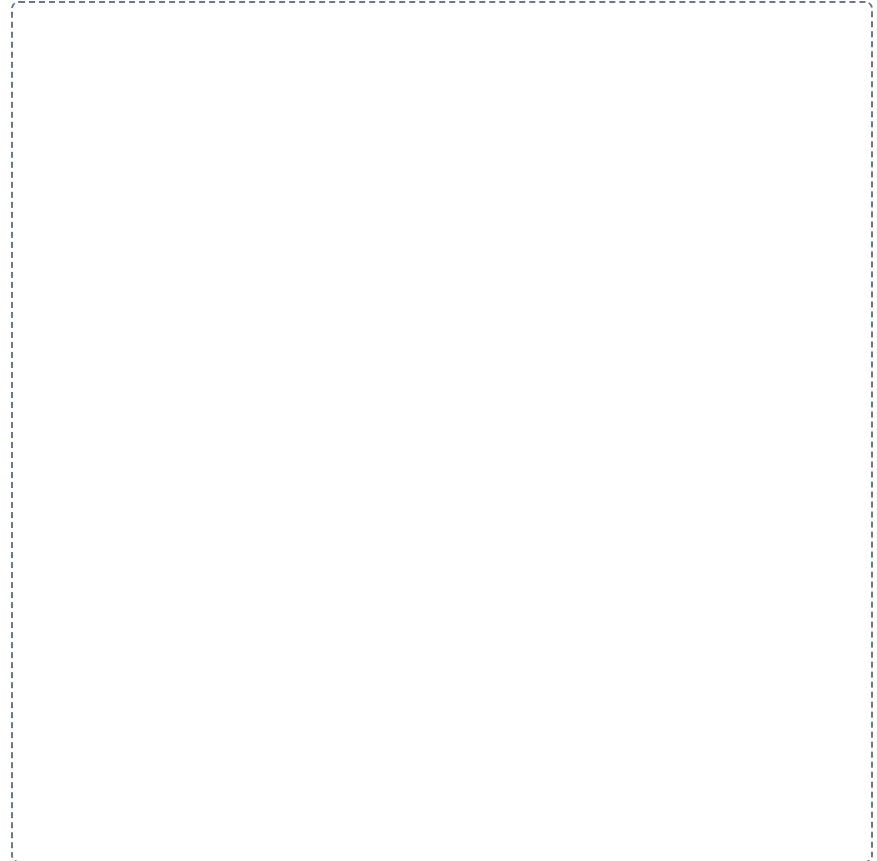


12-4

## The sphenoid

HUB

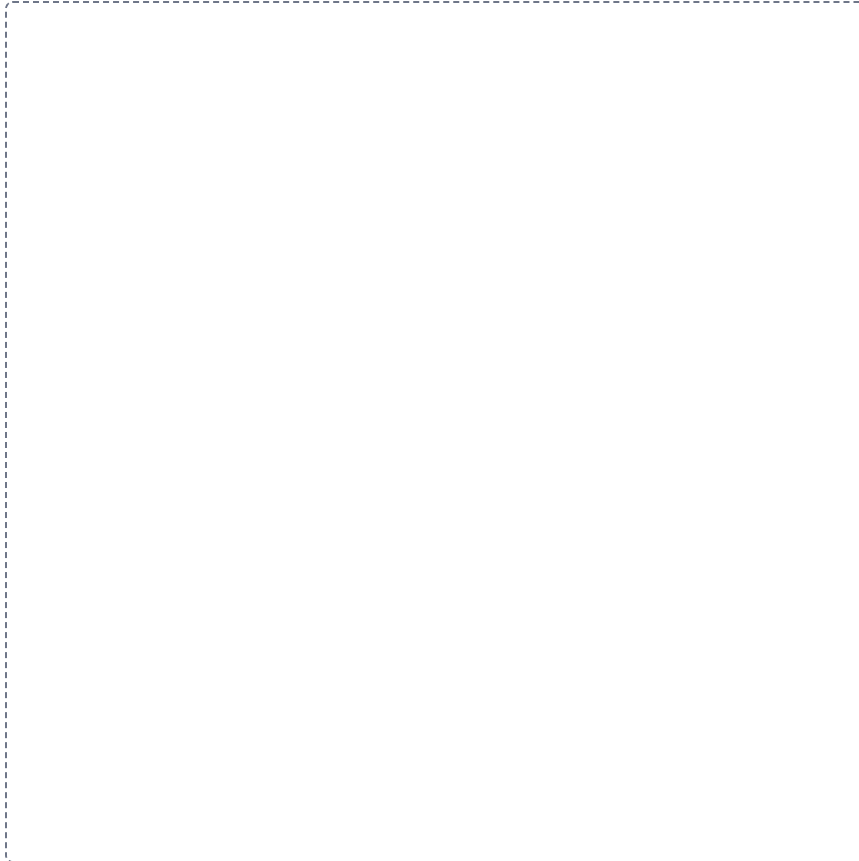
Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



### The four fontanelles

**GRID**

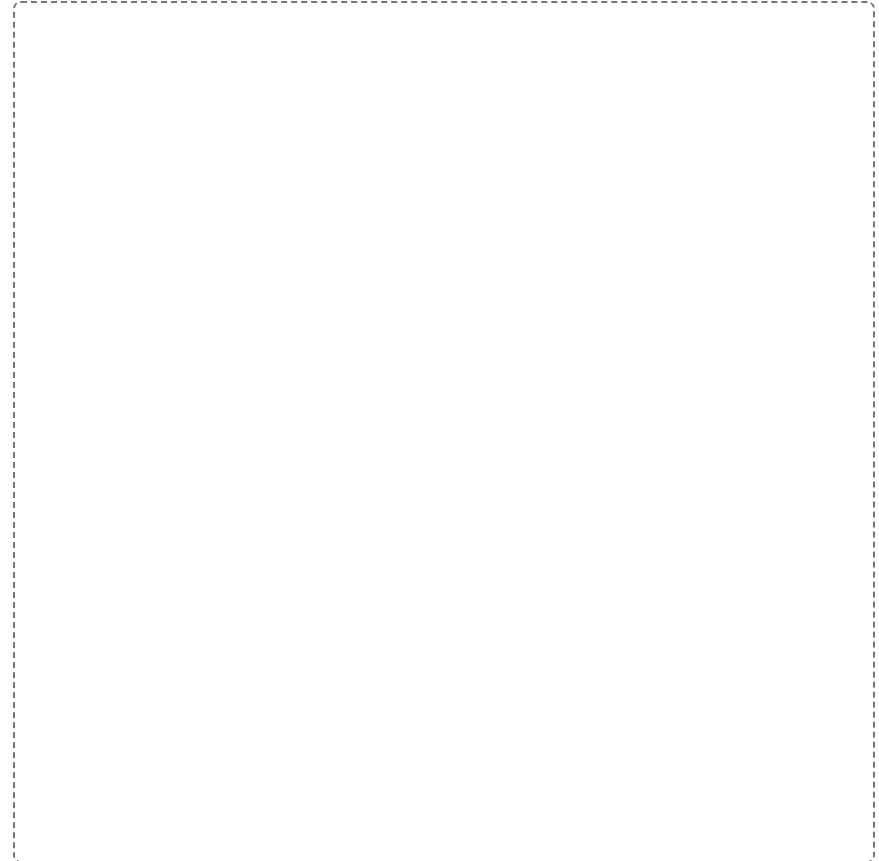
Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



### The five regions of the spine

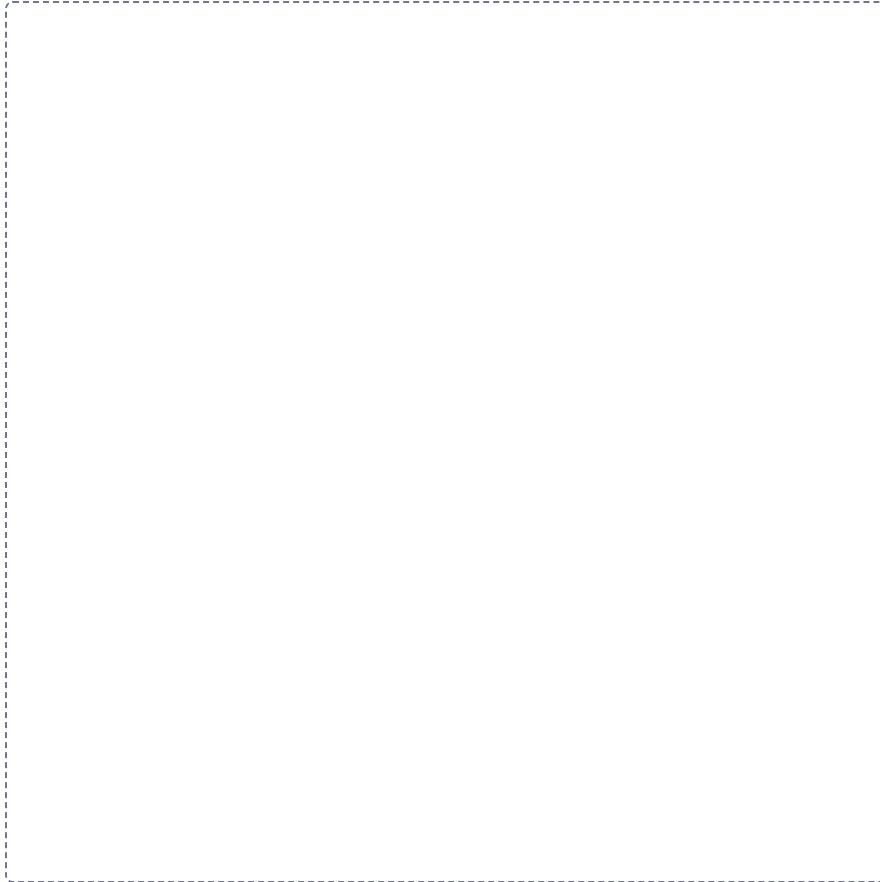
**LADDER**

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.



**Cervical, thoracic, lumbar vertebrae****GRID**

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



**The four curvatures, in developmental order**

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

